FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Emergency Obstetric Care & Safe Motherhood

Lecture | Week 4 · Hypertension, twins & the pelvic floor | Obstetrics | Prof Mirembe

Most maternal deaths come from a few direct complications (haemorrhage, hypertensive disorders, sepsis, obstructed labour, unsafe abortion) that cannot be predicted but can be treated. Safe motherhood therefore means timely access to Emergency Obstetric and Newborn Care (7 BEmONC and 9 CEmONC signal functions), removing the three delays, and learning from every death through MPDSR.

Must know before the ward round

1. Maternal death: death while pregnant or within 42 days of the end of pregnancy, from any cause related to or aggravated by pregnancy or its management, excluding accidental or incidental causes.
2. Uganda MMR fell from 336 (UDHS 2016) to 189 per 100,000 live births (UDHS 2022). SDG 3.1 is a global MMR below 70 by 2030, with no country above 140.
3. BEmONC = 7 signal functions: parenteral antibiotics, uterotonics and anticonvulsants; manual removal of placenta; removal of retained products; assisted vaginal delivery; neonatal resuscitation with bag and mask. CEmONC = these 7 + caesarean section + blood transfusion.
4. A signal function counts only if it was actually performed at least once in the last 3 months.
5. UN standard: at least 5 EmONC facilities, including at least 1 comprehensive, per 500,000 population. Uganda: HC III = BEmONC; HC IV and hospitals = CEmONC.
6. About 15% of pregnant women develop a life-threatening complication. UN targets: met need for EmONC 100%, CS 5–15% of all births, direct obstetric case fatality rate below 1%.
7. Three delays (Thaddeus & Maine 1994): delay 1 in deciding to seek care, delay 2 in reaching a facility, delay 3 in receiving adequate care once there.
8. Rapid initial assessment on arrival, before any history: airway/breathing, shock, heavy bleeding, convulsions or unconsciousness, dangerous fever, severe abdominal pain, imminent delivery. Any 'yes' = treat now.
9. Stabilise before referral: $MgSO_4$ loading dose (4 g IV + 10 g IM), PPH bundle (oxytocin, TXA, fluids) ± NASG, first dose of IV antibiotics, first dose of dexamethasone; escort, note with vitals, drugs and times; phone ahead.
10. MPDSR in Uganda: every maternal death is notifiable within 24 hours and reviewed (ideally within 7 days) in a no-name, no-blame audit that must end in a response (action).
11. Leading causes of maternal death in Uganda: haemorrhage (mostly PPH), hypertensive disorders, sepsis, abortion complications, obstructed labour/ruptured uterus, and indirect causes (malaria, HIV, anaemia, heart disease).
12. Respectful maternity care is a human right and a quality standard: consent, privacy, dignity, a companion of choice, no abuse, no detention for unpaid bills.

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1. Definitions and Key Concepts

The key insight of the safe-motherhood movement is this: **we cannot predict which woman will develop a complication**, because most obstetric emergencies (PPH, eclampsia, obstructed labour, sepsis) occur in "low-risk" women. So the strategy is to make sure **every woman can reach, in time, a facility that can perform the life-saving "signal functions"**. That is **Emergency Obstetric and Newborn Care (EmONC)**.

> **CLINICAL PEARL**
> The one thing never to forget: **"Every pregnancy faces risk."** Prevent what you can (family planning, ANC, skilled birth attendance), but mothers survive because the emergency is **recognised early, reached quickly and treated competently**. That is the three delays turned upside down.

Term	Definition
Maternal death (ICD)	Death of a woman **while pregnant or within 42 days of termination of pregnancy**, irrespective of duration or site of the pregnancy, from any cause related to or aggravated by the pregnancy or its management, but **not** from accidental or incidental causes
Late maternal death	Death from direct or indirect obstetric causes **more than 42 days but less than 1 year** after the end of pregnancy
Pregnancy-related death	Death while pregnant or within 42 days, **irrespective of cause** (includes accidents)
Direct obstetric death	From obstetric complications of pregnancy, labour or puerperium, or from interventions, omissions or incorrect treatment (e.g. PPH, eclampsia, sepsis, ruptured uterus, abortion complications, ectopic, anaesthetic death)
Indirect obstetric death	From a pre-existing disease or one that developed in pregnancy, aggravated by the physiological effects of pregnancy (e.g. **malaria, HIV/TB, anaemia, heart disease**, sickle cell)
Maternal mortality ratio (MMR)	Maternal deaths per **100,000 live births** (measures obstetric risk per pregnancy)
Maternal mortality rate	Maternal deaths per 100,000 **women of reproductive age** (15–49) (reflects both obstetric risk and fertility)
Lifetime risk	Probability that a 15-year-old girl will eventually die from a maternal cause
Maternal near miss	A woman who **nearly died but survived** a complication during pregnancy, childbirth or within 42 days (WHO organ-dysfunction criteria, e.g. shock, need for ≥5 units of blood, hysterectomy for bleeding, eclampsia with coma, ICU admission)
Skilled birth attendant	An accredited health professional (midwife, nurse, doctor, clinical officer) trained to manage normal pregnancy and birth and to identify, manage and refer complications
EmONC	The package of life-saving interventions (signal functions) for obstetric and newborn complications
Signal function	A key medical intervention that treats the major direct causes of maternal death, used as an **indicator** of a facility's capacity (it must have been performed at least once in the last 3 months)

EmONC signal functions (UN, 2009)

#	Signal function	What it treats	BEmONC	CEmONC
1	**Parenteral antibiotics**	Puerperal sepsis, septic abortion, PROM	Yes	Yes
2	**Parenteral uterotonic drugs** (oxytocin)	PPH	Yes	Yes

#	Signal function	What it treats	BEmONC	CEmONC
3	**Parenteral anticonvulsants** ($MgSO_4$)	Pre-eclampsia/eclampsia	Yes	Yes
4	**Manual removal of placenta**	Retained placenta, PPH	Yes	Yes
5	**Removal of retained products** (MVA, D&C)	Incomplete abortion, secondary PPH	Yes	Yes
6	**Assisted vaginal delivery** (vacuum, forceps)	Prolonged 2nd stage, fetal distress	Yes	Yes
7	**Basic neonatal resuscitation** (bag and mask)	Birth asphyxia	Yes	Yes
8	**Surgery (caesarean section)**	Obstructed labour, praevia, abruption, fetal distress, rupture	No	Yes
9	**Blood transfusion**	Haemorrhage, severe anaemia	No	Yes

> NOTE
> A 2025 Delphi consensus (the "Re-Visioning EmONC" project) proposed an **expanded set of 25 signal functions** (11 obstetric, 13 neonatal, 1 referral), adding for example tranexamic acid, antenatal corticosteroids, kangaroo care and referral capacity. For exams, **learn the classic 7 + 2**, and mention that the framework is being updated.

2. Burden and Uganda Context

Each year about **260,000 women die** from pregnancy and childbirth worldwide, and about **70% of them in sub-Saharan Africa**. In Uganda roughly **1 in every 530 live births** is accompanied by a maternal death (MMR 189/100,000, UDHS 2022). Most deaths happen **around labour, delivery and the first 24 hours after birth**.

Global picture

- About **260,000 maternal deaths in 2023** (global MMR about 197); **about 70% in sub-Saharan Africa**. Most would be preventable with timely, quality care.
- **SDG 3.1**: reduce the global MMR to **<70 per 100,000 live births by 2030**, with **no country above 140**. **SDG 3.1.2**: proportion of births attended by skilled personnel.
- The **Millennium Development Goal 5** (reduce MMR by 75% between 1990 and 2015) was missed by most of sub-Saharan Africa, including Uganda.

Uganda: trends

Survey	MMR (per 100,000 live births)
UDHS 2000–01	505
UDHS 2006	435
UDHS 2011	438
UDHS 2016	**336**
UDHS 2022	**189**

This is real progress, driven by more facility births (**about 9 in 10 births now occur in a health facility** in UDHS 2022), more CEmONC at HC IVs, MPDSR and results-based financing. But **189 is still far above the SDG target**, and the **quality of care** at facilities, not just access, is now the main battleground (the "**third delay**").

Causes of maternal death in Uganda (MPDSR reports and facility studies)

Cause	Comment
Obstetric haemorrhage (especially **PPH**)	Leading cause nationally, around a quarter to a third of deaths
Hypertensive disorders (pre-eclampsia/eclampsia)	Often second; linked to late or low-quality ANC and failure to give $MgSO_4$
Sepsis (puerperal, post-abortion, post-CS)	Leading cause in some tertiary series (about 31% in one 2016 series **Verify** *that series appears to come from Mbarara Regional Referral Hospital, not Mulago as previously stated*)
Abortion complications	Unsafe abortion; restrictive law and stigma delay care
Obstructed labour / ruptured uterus	Late referral, herbal uterotonics
Indirect causes	**Malaria, HIV/TB, anaemia**, cardiac disease, sickle cell disease
Anaesthetic complications	Spinal/GA without trained anaesthetists at lower levels

The Uganda health system for mothers

Level	Catchment (approx.)	Maternity role
VHT (village health team)	Village	Health education, birth preparedness, identifying danger signs, linking to facilities; community misoprostol in some districts
HC II	Parish (~5,000)	ANC, FP, outpatient; **not meant for deliveries** (but many happen)
HC III	Sub-county (~20,000)	Normal deliveries, **BEmONC**, maternity ward, laboratory
HC IV	County / health sub-district (~100,000)	**CEmONC**: theatre (CS), blood transfusion, doctor(s)
General (district) hospital	District (~500,000)	CEmONC, more specialists
Regional referral hospital (RRH)	Region (~2 million)	Obstetricians, HDU, blood bank, internship training
National referral hospitals	National	e.g. **Kawempe NRH** (dedicated maternity), Mulago Specialised Women and Neonatal Hospital; ICU; subspecialists
PNFP/private hospitals	—	e.g. **Mengo** (Church of Uganda), Nsambya, Rubaga, Lubaga: about a third of services

UGANDA CONTEXT

Common gaps at HC IVs, which you will see on outreach and in referral letters: no anaesthetist or a non-functional theatre, **no blood**, oxytocin or $MgSO_4$ stock-outs, one midwife for the whole maternity at night, no ambulance fuel, and referral forms without vital signs. Knowing these realities helps you understand the "third delay" and write better referral and MPDSR reports.

3. Principles and Framework

Complication: about 15% of all pregnancies, often in low-risk women

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Delay 1: deciding to seek care

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Delay 2: reaching a facility

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Delay 3: receiving adequate care

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Signal function given in time: survival | Signal function missing or late: near miss or death

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MPDSR: notify, review, respond

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Fix the delay: education, transport, staffing, supplies, drills

↻ From “Fix the delay: education, transport, staffing, supplies, drills” back to “Delay 1: deciding to seek care”

The three delays model (Thaddeus & Maine, 1994, *"Too far to walk"*)

Delay	What it is	Examples of causes (Uganda)	Interventions
1. Deciding to seek care	Woman/family doesn't recognise danger or decides not to go	Not knowing danger signs; husband or mother-in-law decides; fear of costs, of CS, of abusive staff; belief in TBAs or herbalists; previous bad experience; stigma (abortion, HIV)	Health education, **birth preparedness & complication readiness**, male involvement, VHTs, **respectful care**, community dialogues, free services
2. Reaching a facility	Physically getting to care	Distance, rough roads, no transport at night, cost of boda-boda, floods, insecurity, being referred between facilities	Maternity waiting homes, community transport schemes, ambulance networks, vouchers, **HC IV CEmONC** closer to people, mobile phone referral
3. Receiving adequate care	Care is delayed or poor at the facility	Staff shortage, lack of drugs/blood/theatre, poor triage, delayed decisions, poor skills, **referral without stabilisation**, demand for payment or supplies ("buy gloves")	Staffing, supplies, **EmONC training** (e.g. drills), protocols, triage, MPDSR, supportive supervision, quality improvement

EXAM FAVOURITE

Examiners frequently ask you to **analyse a maternal death using the three delays**. Structure your answer delay by delay, and within each delay name **patient/family, community, and health-system** factors. Then propose **one action per delay**.

Pillars of Safe Motherhood

The **Safe Motherhood Initiative** was launched in **Nairobi in 1987** (WHO, UNFPA, UNICEF, World Bank). The classic **four pillars**, resting on a base of **primary health care and equity for women** (education, status, empowerment):

1. **Family planning**: fewer unwanted, high-risk, and closely spaced pregnancies; fewer unsafe abortions.
2. **Antenatal care**: screening and prevention (iron/folate, IPTp, TT, HIV/syphilis testing, BP), and birth preparedness.

3. **Clean and safe delivery**: a skilled attendant at every birth.
4. **Essential (emergency) obstetric care**: EmONC for complications.

Later frameworks add **postnatal care**, **post-abortion care**, prevention of STIs/HIV, and **respectful maternity care**.

Why skilled attendance plus EmONC, not "risk screening"?

- Risk scoring at ANC (age, parity, height) **misses most women who develop complications**: "low-risk" women still bleed and convulse.
- About **15% of all pregnant women develop a potentially life-threatening complication**, and about 5–15% of births need a CS. So we plan services for that 15%, for everyone.
- **CS rate as a population indicator**: <5% suggests unmet need (women dying from obstructed labour); >15% suggests possible over-use. Uganda's national CS rate is below this band in rural areas but much higher in urban private hospitals: both an access and an equity problem.

UN EmONC process indicators (2009 handbook)

1. **Availability**: ≥5 EmONC facilities (≥1 comprehensive) per 500,000 population.
2. **Geographic distribution**: all sub-national areas meet indicator 1.
3. **Proportion of all births in EmONC facilities**.
4. **Met need for EmONC**: proportion of women with major direct obstetric complications treated in EmONC facilities (estimated as 15% of expected births); target **100%**.
5. **CS as a proportion of all births**: 5–15%.
6. **Direct obstetric case fatality rate**: **<1%**.
7. **Intrapartum and very early neonatal death rate**.
8. **Proportion of maternal deaths due to indirect causes** in EmONC facilities.

MPDSR: Maternal and Perinatal Death Surveillance and Response

- Uganda made **maternal and perinatal deaths notifiable** (2008), with national **MPDSR guidelines (2017)**.
- **Cycle**: **identify and notify** (within 24 h, via the facility and DHIS2/weekly surveillance) → **review** (facility MPDSR committee, ideally within 7 days; case notes, staff interviews, **three-delays analysis**, cause of death using ICD-MM) → **analyse** (avoidable factors, patterns) → **respond** (action plan with responsible person and deadline) → **monitor and evaluate** the response.
- **Principles**: **no name, no blame**; confidentiality; the goal is learning, not punishment; the review isn't complete without a **response**.
- **Community**: **verbal autopsy** for deaths outside facilities.
- **Perinatal deaths** (stillbirths and early neonatal deaths) are reviewed in the same system.
- Uganda produces a **National Annual Maternal and Perinatal Death Report**. Common findings: late referral, delayed CS, lack of blood, and failure to monitor after delivery.

Respectful maternity care (RMC)

The WHO **"positive childbirth experience"** (2018 intrapartum recommendations) and the White Ribbon Alliance **Respectful Maternity Care Charter** state women's rights to:

- freedom from harm and ill-treatment; information, **informed consent** and refusal; **privacy and confidentiality**; **dignity and respect**; equality and non-discrimination (HIV status, age, poverty, disability); timely care; **liberty** (no detention for unpaid bills); a **companion of choice** during labour.

Seven categories of **disrespect and abuse** (Bowser & Hill): physical abuse, non-consented care, non-confidential care, non-dignified care (shouting, scolding), discrimination, **abandonment or neglect** (delivering alone), and **detention in facilities**.

Why it matters clinically: disrespect drives the **first delay**. Women who were slapped or shouted at last time deliver at home or with a TBA next time.

4. Application in Practice

In this topic the "workup" has two faces: **(A) the systematic assessment of any woman arriving with an obstetric emergency**, which ends in the right **signal function** at the right level, and **(B) the systematic review of a maternal death**, which ends in a **response**. Steps 0–5 follow the woman; the maternal death review is worked through in Step 5.

Step 0: First 5 minutes

Rapid initial assessment (RIA) at the door, done by whoever meets her first (WHO MCPC approach), in under a minute:

Look for	If present	Immediate action
Airway/breathing: very difficult breathing, cyanosis	Emergency	Airway, oxygen, sit up (if pulmonary oedema), call for help
Shock: cold clammy skin, pulse >110, SBP <90, confusion; **shock index ≥0.9**	Emergency	Lie flat/left tilt, 2 large-bore cannulae, warmed fluids, find the bleeding
Heavy vaginal bleeding	Emergency	PPH bundle or APH pathway (no VE in APH)
Convulsing or unconscious	Emergency	Recovery position, airway, **$MgSO_4$ loading dose**, check BP and glucose
Dangerous fever (≥38 °C with weakness, stiff neck, confusion)	Emergency	Cultures if possible, **first dose of IV antibiotics**, malaria test
Severe abdominal pain	Emergency	Think rupture, abruption, ectopic, sepsis
Imminent delivery (urge to push, head visible)	Emergency	Deliver; prepare for newborn resuscitation
Priority signs, e.g. labour (including prolonged labour), ruptured membranes, severe headache or blurred vision, severe pallor, vomiting	Priority	Full assessment next; do not leave her in the queue

Anyone with a "yes" gets **emergency care immediately**. Use **obstetric early warning scores (MEOWS)** on the wards to catch deterioration.

Then, in the same 5 minutes:

1. **Shout for help** and assign roles; note the **time of arrival**.
2. **Full set of vitals**: pulse, BP, RR, SpO_2, temperature, conscious level; start a **MEOWS/early warning chart**.
3. **IV access and bloods** (Hb, group and cross-match, clotting tube); **catheter** if shocked, eclamptic or in obstructed labour.
4. **Name the emergency and the signal function it needs** (table below), then decide: **can this facility do it now?** If not, stabilise and refer (see section 6B).

Emergency	Signal function(s) needed	HC III (BEmONC)	HC IV / hospital (CEmONC)
PPH (atony)	Parenteral uterotonic (+ TXA, fluids)	Yes	Yes
Retained placenta	Manual removal of placenta	Yes	Yes
Incomplete abortion / retained products	Removal of retained products (MVA)	Yes	Yes
Severe pre-eclampsia / eclampsia	Parenteral anticonvulsant ($MgSO_4$)	Yes, then refer for delivery if needed	Yes
Sepsis	Parenteral antibiotics	Yes (first dose, then refer if source control needed)	Yes
Prolonged 2nd stage, fetal distress (criteria met)	Assisted vaginal delivery	Yes	Yes
Non-breathing newborn	Neonatal resuscitation (bag and mask)	Yes	Yes
Obstructed labour, praevia, abruption with live fetus, rupture	**Caesarean section / laparotomy**	**No: stabilise and refer**	Yes
Massive haemorrhage, severe anaemia	**Blood transfusion**	**No: stabilise and refer**	Yes

Step 1: Focused history

Danger signs to ask every pregnant or postnatal woman

Danger sign	Think of
Vaginal bleeding (any amount in pregnancy; heavy after birth)	Abortion, ectopic, APH, PPH
Convulsions / fits, severe headache, blurred vision, epigastric pain	Pre-eclampsia/eclampsia (also malaria, epilepsy, meningitis)
Fever, foul-smelling discharge, abdominal pain	Sepsis, malaria, chorioamnionitis
Severe abdominal pain	Abruption, rupture, ectopic, sepsis, surgical causes
Labour >12 hours, or contractions without progress	Obstructed/prolonged labour
Draining liquor before labour	PROM/PPROM
Reduced or absent fetal movements	Fetal compromise/IUFD
Difficulty breathing, fast heartbeat, extreme weakness	Anaemia, heart failure, PE, sepsis
Swelling of face and hands	Pre-eclampsia

For a woman who has been referred, **also ask the three delays questions**: *When did the problem start? When did you decide to come? How did you travel, and how long did it take? Where did you go first, and what was done there?* This is both clinically useful (duration of labour, doses already given) and good MPDSR data.

For the woman in front of you (and her referral note)

Ask the woman, her companion and **read the referral note and ANC card**. Each item changes what you do:

- [ ] **What is the problem, and when did it start?** → duration of labour, bleeding or fits; TXA works only within 3 h of birth.
- [ ] **Gestation, parity, time of birth (if delivered)** → antepartum vs intrapartum vs postpartum pathway; viability; steroids.

- [ ] **Drugs already given, doses and times** (referral note) → avoid double-loading $MgSO_4$; know when the next dose is due (e.g. 5 g IM 4 h after loading); count uterotonic doses.
- [ ] **Previous CS or uterine surgery** → rupture, praevia/PAS.
- [ ] **ANC card**: BP, urine protein, Hb, **HIV status and ART**, syphilis, blood group, number of visits → pre-eclampsia, anaemia, eMTCT, anti-D.
- [ ] **Where did she go first, and what was done?** (TBA, herbalist, HC II/III, private clinic) → herbal uterotonics, fundal pressure, unsafe abortion; also delay data.
- [ ] **The three-delays questions**: when did she decide to come? how did she travel, how long did it take? how long did she wait here? → part of the clinical story and of any MPDSR or near-miss review.
- [ ] **Chronic illness**: heart disease, sickle cell, epilepsy, asthma, diabetes → indirect causes; contraindications (ergometrine, carboprost).
- [ ] **Consent and next of kin**; who can donate blood → you will need both quickly.

Step 2: Focused examination

Follow the order of Hutchison's Clinical Methods (24th edition), but **only after the rapid initial assessment and resuscitation** (Step 0). In an emergency, examine while the team resuscitates and complete the sequence once she is stable.

Introduction, consent and preparation

- Introduce yourself to the woman and her companion; explain what you are doing and obtain **consent** (if she cannot consent, act in her best interests and inform the next of kin).
- Privacy, a **chaperone**, **hand hygiene** and gloves.
- **Position**: left lateral tilt from 20 weeks (or the recovery position if convulsing or unconscious); expose only what you need.

General inspection and vital signs

General: conscious level (AVPU/GCS), pallor, jaundice, oedema, dehydration, signs of respiratory distress.

Vital signs with trend: pulse, BP, RR, SpO_2, temperature, **shock index**, urine output (≥30 mL/h).

Hands, face and neck

- **Hands**: cold, clammy peripheries and slow capillary refill (shock); pallor of the palms (anaemia).
- **Face**: conjunctival pallor, jaundice, central cyanosis, facial oedema (pre-eclampsia).
- **Neck**: neck stiffness (meningitis, cerebral malaria); raised JVP (heart failure, fluid overload).

System checks by presenting problem

- **Chest and heart**: crackles (pulmonary oedema, pneumonia), murmur (cardiac disease).
- **Neuro**: GCS, neck stiffness, focal signs, reflexes ($MgSO_4$ monitoring).

Obstetric examination

In the order of the obstetric abdominal examination (inspection, fundal height, lie, presentation, engagement in fifths, fetal heart), then the postpartum, vaginal and newborn checks:

- **Abdomen**: fundal height, uterine tone and tenderness, scar tenderness, **Bandl's ring**, fetal lie and presentation, descent (fifths), contractions, **fetal heart**.
- **Postpartum**: fundus (firm? high?), lochia, perineum, wound.
- **Vaginal examination** only if there is **no APH** (or praevia has been excluded): dilatation, station, moulding, caput, liquor colour; in obstructed labour, look for oedema of the cervix and vulva.
- **Newborn**, if delivered: breathing, heart rate, temperature, feeding.

Completing the examination

- Cover her, thank her and her companion, **hand hygiene**.
- **Name the emergency and the signal function** it needs, and say whether this facility can deliver it now.

Document these positives	And these negatives
Time of arrival and time of each action	"No VE (APH, praevia not excluded)" when relevant
Vitals with shock index; GCS	"No signs of rupture", "fetal heart present"
Emergency named and signal function given	"No $MgSO_4$ toxicity: RR ≥16, reflexes present, urine ≥30 mL/h"
Decision (treat here or refer) and the reason	"Referral note checked: drugs and times reconciled"

Step 3: Bedside tests

Test	When	What it tells you
Shock index, calibrated drape/pad weighing	Any bleeding	Severity; objective PPH trigger
HemoCue / point-of-care Hb	Bleeding, pallor, before referral	Need for blood (and therefore CEmONC)
20-minute whole blood clotting test	Bleeding, abruption, IUFD, sepsis, eclampsia/HELLP	Coagulopathy without a lab
Urine dipstick (protein)	Every pregnant woman with headache, fits, high BP	Pre-eclampsia
Random blood glucose	Convulsing, unconscious or confused	Hypoglycaemia (severe malaria, sepsis)
Malaria RDT	Fever, coma, anaemia	Severe malaria, an indirect cause
HIV rapid test (if status unknown), syphilis test	All admissions without a recent result	eMTCT, co-trimoxazole, infant prophylaxis
Urine pregnancy test	Collapse or abdominal pain in any woman of reproductive age	Ruptured ectopic
Fetal heart by Doppler/Pinard (CTG in hospital)	Antepartum and intrapartum	Viability, fetal distress
Partograph / Labour Care Guide review	Every woman in labour or referred in labour	Alert/action line crossed, obstruction
Point-of-care ultrasound (skilled operator)	APH, collapse, suspected rupture/ectopic	Placental site, fetal heart, free fluid

Step 4: Laboratory & imaging

Test	When	Expected / abnormal finding	How it changes management
Full blood count	All emergencies	Hb <7 g/dL severe anaemia; platelets <100 × 10^9/L; high or very low WBC	Transfusion (a CEmONC signal function); HELLP/DIC; sepsis. HC IV and above
Group and cross-match	Bleeding, anaemia, planned CS	Compatible units	Only where blood is stocked (HC IV with a fridge, hospitals via UBTS); otherwise **refer with a potential donor**
PT/APTT, fibrinogen	Major haemorrhage, abruption, sepsis, AFE	Prolonged; fibrinogen <2 g/L	FFP, cryoprecipitate. RRH/NRH only

Test	When	Expected / abnormal finding	How it changes management
Creatinine, urea, electrolytes	Pre-eclampsia, sepsis, shock, obstructed labour	Raised creatinine, high K^+	AKI: fluid balance, drug doses (gentamicin, $MgSO_4$); refer for HDU
LFTs	Pre-eclampsia, jaundice	Raised transaminases, bilirubin	HELLP, hepatitis, AFLP; expedite delivery
Blood cultures, HVS, urine culture	Sepsis	Organism and sensitivity	Adjust antibiotics. RRH/NRH (limited)
Blood smear for malaria	Fever, coma	Parasitaemia	IV artesunate
Lactate / blood gas	Septic or haemorrhagic shock	Lactate >2 (sepsis), >4 (severe) mmol/L	Severity; ICU referral. NRH/ICU only
Obstetric ultrasound	APH, suspected rupture, IUFD, retained products, ectopic	Placental site, fetal viability, free fluid, retained tissue	Mode and place of delivery or surgery. HC IV (if sonographer) and above
Chest X-ray, ECG, echo	Breathlessness, suspected cardiac disease, PE	Pulmonary oedema, cardiomegaly, arrhythmia	Cardiology/ICU; hospital level
CT head	Focal signs, persistent coma after eclampsia	Stroke, haemorrhage	NRH/private only

Step 5: Putting it together

Model one-line summary (emergency)

> "A 19-year-old primigravida at 38 weeks, referred from a HC III 3 hours away with eclampsia (two fits), loaded with $MgSO_4$ at 14:00, now BP 172/114, GCS 13, reflexes present, in early labour with a live fetus: eclampsia with severe hypertension, needing continued $MgSO_4$, acute BP control and delivery at a CEmONC facility."

Problem list

1. Eclampsia on $MgSO_4$ (next IM dose due 18:00).
2. Severe hypertension (≥160/110) not yet treated.
3. Delivery needed (augment or CS).
4. Screen for HELLP, AKI, pulmonary oedema.
5. Delays 1 and 2 documented for near-miss review.

Working diagnosis: *eclampsia with severe-range BP at term; signal functions needed: parenteral anticonvulsant (given), CS available if needed.*

EXAM FAVOURITE

Worked example 1: triage of four women arriving together at a HC IV. (a) Para 3, 1 h after home birth, soaked wrapper, pulse 128, BP 96/60 → **SI 1.3**: **emergency**, first. Bundle now (massage, oxytocin, TXA, fluids, examination). (b) Primigravida at 36 weeks, convulsing: **emergency**, simultaneously (second team): airway, **$MgSO_4$ 4 g IV + 10 g IM**, then BP. (c) Para 1, day 5 postpartum, 38.6 °C, walking, alert, pulse 104: **priority**. Assess next for puerperal sepsis; first dose of IV antibiotics within the hour. (d) Para 2 in labour, 5 cm, FHR 140, comfortable: **non-urgent**, admit and start a partograph. Staff are allocated by these decisions, not by order of arrival.

EXAM FAVOURITE

Worked example 2: a maternal death review (MPDSR), step by step. *Summary of events*: 24-year-old para 4 delivered at a HC III at 22:00; AMTSL given; not observed overnight; found at 06:00 pale and bleeding; referred at 08:00 without an IV line; died on arrival at the HC IV at 10:30.

1. **Cause of death**: underlying cause **PPH due to uterine atony** (direct obstetric death); contributing factor: **antenatal anaemia** (Hb 8.9 g/dL, not treated).
2. **Delays**: delay 1: none (she delivered in a facility). Delay 2: 2.5 h transfer; no ambulance fuel. Delay 3: **no postpartum monitoring for 8 hours**, no bundle, referral without stabilisation.
3. **Avoidable factors**: health worker (no observation, no IV), facility (one midwife at night, no calibrated drapes), system (ambulance), patient/community (none identified).
4. **Response (SMART)**: "Observations every 15 min for the first 2 h after birth, then 30-minutely to 6 h, recorded on a postnatal chart; audited weekly by the in-charge from next month"; "district to ring-fence ambulance fuel by the end of the quarter"; "PPH drill for all HC III staff within 6 weeks".
5. **Notify within 24 h**, review within 7 days, **no name, no blame**, follow up that the actions happened.

EXAM FAVOURITE

Worked example 3: EmONC indicators for a district. (Illustrative numbers.)

- Expected births 40,000 → expected women with major direct complications = 15% = **6,000**. Treated in EmONC facilities: 2,700 → **met need = 45%** (target 100%).
- CS 1,200 / 40,000 births = **3%** → below 5%: **unmet need** for surgery.
- Direct obstetric deaths in EmONC facilities 40 / women with direct complications treated 2,700 = **1.5%** → above the **<1%** target: look at the **quality of care** (third delay).

Model presentation of a referred emergency

> "Mrs L, a 19-year-old primigravida at 38 weeks, referred from a HC III 40 km away with eclampsia. She fitted twice at home, and the family delayed 6 hours before seeking care (first delay: they consulted a herbalist first). Transport took 3 hours on a boda-boda then an ambulance (second delay). She received $MgSO_4$ 4 g IV and 10 g IM at the HC III at 14:00, but no antihypertensive, and no BP is recorded on the referral note. On arrival: BP 172/114, GCS 13, reflexes present, RR 18, urine protein 3+, cephalic, 4 cm, FHR 150. Plan: continue $MgSO_4$ maintenance, hydralazine 5 mg IV, bloods (FBC, LFTs, creatinine, clotting), catheter, deliver within 12 hours (augment or CS), and note the delays for the MPDSR/near-miss audit."

5. Clinical Acumen

This topic is about systems, but your acumen is still tested on the ward: **recognising the emergency, triaging it and referring it correctly**.

Thinking like a clinician (and like a public-health doctor)

- **Every maternal death has a story**: analyse it by delays and by missed opportunities (ANC, labour monitoring, postpartum observation).
- **"Stabilise before transfer"**: pre-referral treatment saves lives. A woman with eclampsia should get her **$MgSO_4$ loading dose** before the ambulance; PPH needs **oxytocin, TXA, fluids and NASG**; sepsis needs the **first antibiotic dose**.
- **The first 24 hours postpartum** is the most dangerous period. Keep women at the facility for **at least 24 hours** after birth.

- **Indirect causes** kill Ugandan mothers too: test for **malaria and HIV**, look for **anaemia**, and ask about heart disease.
- **Pitfall**: thinking of EmONC as "buildings and equipment". The signal function must actually have been **performed** in the last 3 months to count; a vacuum extractor in a cupboard is not assisted vaginal delivery.

Red flags in the system

RED FLAG

- A **referral note without vital signs, drugs given or times**: re-assess everything
- A woman **referred in shock without an IV line** or without $MgSO_4$ for eclampsia
- **Decision-to-delivery interval** for an emergency CS of more than 30–60 minutes
- **No monitoring after birth** (first 2 hours unobserved), where most PPH deaths happen
- **Stock-outs** of oxytocin, $MgSO_4$, antibiotics, blood; theatre closed at night
- A mother "waiting for relatives to buy supplies" before surgery

Scenarios: the cause of collapse or death around childbirth

When a woman collapses or dies around childbirth, the differential of the **cause** guides both care and the MPDSR review.

Condition	Distinguishing features	Key investigation
PPH / APH (hypovolaemic shock)	Visible or concealed bleeding, soft uterus or known praevia/abruption	Clinical; Hb; clotting test
Eclampsia / stroke	Seizures, BP ≥160/110, proteinuria, headache	BP, urine protein, (CT if focal signs)
Sepsis	Fever or hypothermia, tachycardia, foul discharge, post-abortion or post-CS	FBC, cultures, malaria test, lactate
Ruptured uterus	Obstructed labour or scar, shock, fetal parts palpable	Clinical, laparotomy
Amniotic fluid embolism	Sudden collapse in labour or just after, hypoxia, DIC	Clinical diagnosis
Pulmonary embolism	Sudden dyspnoea, chest pain, often postpartum or post-CS	Clinical ± CTPA/echo where available
Severe malaria	Fever, coma, hypoglycaemia, anaemia	RDT/smear, glucose
Peripartum cardiomyopathy / cardiac disease	Orthopnoea, oedema, murmur, late pregnancy to 5 months postpartum	Echo, CXR, ECG
Anaesthetic complication	High spinal, failed intubation, aspiration	Clinical history

Common student mistakes

- Triaging by order of arrival instead of by **danger signs**.
- Referring without **stabilising** (no IV line, no $MgSO_4$ loading dose, no first antibiotic dose).
- Writing a referral note without **vital signs, drugs, doses and times**.
- Counting equipment as a signal function when it was not **used in the last 3 months**.
- Calling an accidental death in pregnancy a "maternal death", or a malaria death 3 weeks postpartum "not maternal".
- Turning an MPDSR review into a search for someone to blame, or ending it without a **response**.

6. Common Problems and Their Management

A. At the individual level: managing an obstetric emergency (the EmONC approach)

1. **Rapid initial assessment and triage** on arrival (section 4, Step 0). Shout for help.
2. **Resuscitate** (ABC): airway, oxygen, **IV access with two large-bore cannulae**, fluids, left lateral tilt if pregnant, keep warm, catheter.
3. **Identify the problem** quickly: bleeding, fits, fever, prolonged labour, collapse.
4. **Give the signal-function treatment** without delay:

Emergency	First-line EmONC treatment (dose)
PPH	Massage; **oxytocin 10 IU IV/IM** + infusion (20 IU in 1 L at 60 drops/min); **TXA 1 g IV over 10 min**; fluids; examine; **misoprostol 800 µg SL** if no oxytocin; MRP if retained placenta; bimanual compression; NASG
Severe pre-eclampsia / eclampsia	**$MgSO_4$**: loading **4 g IV** (20% solution over 5–10 min) **+ 10 g IM** (5 g deep IM in each buttock, 50% solution with 1 mL of 2% lidocaine), then **5 g IM every 4 h** in alternate buttocks for **24 h after delivery or the last fit** (Pritchard, used in Uganda); check reflexes, RR ≥16, urine ≥30 mL/h. Antidote: **calcium gluconate 1 g IV** (10 mL of 10%). BP ≥160/110: **hydralazine 5 mg IV** slowly, repeat every 20–30 min (max 20 mg), or **nifedipine 10 mg orally** (repeat after 30 min), or labetalol
Sepsis (puerperal, septic abortion, chorioamnionitis)	**Ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV daily + metronidazole 500 mg IV 8-hourly**; fluids; find and remove the source (evacuate retained products, drain pus)
Incomplete abortion	**MVA** (or misoprostol 400–600 µg as per protocol) + antibiotics if infected; post-abortion family planning
Prolonged 2nd stage / fetal distress	**Vacuum extraction** if criteria are met; otherwise CS
Obstructed labour / praevia / rupture	**Caesarean section or laparotomy** (CEmONC); catheter; antibiotics; blood
Birth asphyxia	**Helping Babies Breathe**: dry, stimulate, clear airway if needed, **bag-and-mask ventilation within the Golden Minute**

5. **Decide: treat here or refer?** If the needed signal function isn't available (e.g. CS or blood at a HC III), **stabilise and refer**.
6. **Monitor and document** (time-stamped), and **communicate** with the family.

B. Referral: step by step

> **UGANDA CONTEXT**
> Uganda's referral chain is **VHT → HC II → HC III → HC IV → General hospital → RRH → NRH**, but **emergencies should go directly to the nearest facility that can perform the needed signal function**, skipping levels where possible. Phone ahead, and use the district ambulance system where available.

1. **Decide early.** Refer when it's indicated, not when the woman is moribund.
2. **Explain** to the woman and family: why, where, the cost, and who should come (a potential **blood donor**).
3. **Stabilise before and during transfer:**

- IV line with fluids running (two lines if bleeding)
- **Eclampsia**: $MgSO_4$ loading dose given (4 g IV + 10 g IM), BP controlled
- **PPH**: oxytocin infusion, TXA, NASG, bimanual/aortic compression during transfer if needed
- **Sepsis**: first dose of IV antibiotics
- **Preterm labour**: first dose of dexamethasone; baby in **kangaroo** position if born
- **Obstructed labour**: IV fluids, **catheter**, antibiotics, no oxytocin

4. **Referral note**: identity; history; findings **with vital signs and times**; **drugs given (dose and time)**; reason for referral; name of the referring worker and a phone number. Include the partograph.
5. **Transport**: ambulance where possible; **accompanied by a health worker** who can manage complications en route, with an emergency kit (oxytocin, $MgSO_4$, gloves, delivery kit, bag and mask).
6. **Communicate** with the receiving facility (phone) so they can prepare theatre and blood.
7. **Counter-referral / feedback** from the receiving facility, which closes the loop and improves care.

C. At the facility level: making EmONC work

- **Staffing**: a skilled birth attendant for every birth, 24/7; an anaesthetic officer and a doctor at every HC IV.
- **Essential supplies**: oxytocin (cold chain), misoprostol, $MgSO_4$, calcium gluconate, antibiotics, TXA, IV fluids, MVA kits, vacuum extractors, NASG, blood (UBTS supply chain).
- **Protocols and drills**: PPH, eclampsia and neonatal resuscitation drills (e.g. **ALSO, PRONTO, HBB, Emergency Obstetric Care skills training**).
- **Triage** and **early warning scores**; 24-hour postnatal stay; functional theatre.
- **Quality improvement**: MPDSR, near-miss reviews, criterion-based audits, clinical mentorship, supportive supervision.
- **Respectful care**: privacy screens, companions of choice, informed consent, no informal payments.

D. At the community and population level: Safe Motherhood interventions

Stage	Interventions (Uganda examples)
Before pregnancy	Girls' education, delaying marriage, **family planning** (unmet need about 1 in 5 married women), nutrition, HIV prevention
Pregnancy	**At least 8 ANC contacts** (WHO 2016; Uganda adopting), first contact in the **first trimester**; iron/folate; IPTp-SP; LLIN; TT/Td; HIV/syphilis/hepatitis B testing (eMTCT); BP and urine; **birth preparedness** (plan place of birth, transport, money, blood donor, companion, **mama kit**)
Labour and birth	**Skilled birth attendance** in a facility, **partograph/Labour Care Guide**, AMTSL, clean birth, EmONC
Postnatal	Stay ≥24 h; postnatal checks (within 24 h, day 3, days 7–14, 6 weeks); FP; danger-sign education
Post-abortion care	Emergency treatment (MVA), counselling, FP, links to other services
Health system	Maternity waiting homes; free delivery at public facilities; results-based financing and vouchers; community transport; MPDSR; data use (DHIS2)

E. Follow-up: MPDSR in practice (what you do after a maternal death)

1. **Notify** within **24 hours** (phone and form) to the district and MoH.
2. **Collect** the file, partograph, drug charts, referral notes; interview staff and relatives (with sensitivity).

3. **Convene the facility MPDSR committee** (ideally within 7 days): present a **summary of events** chronologically; agree the **cause of death** (underlying cause, contributing factors); identify **delays and avoidable factors** at patient, community and facility levels.
4. **Recommend actions** (SMART: specific, measurable, achievable, relevant, time-bound), each with a responsible person.
5. **Implement and follow up**; report to the district and the national committee.
6. **Keep it no-name, no-blame**: focus on systems, not individuals.

F. Judging a service: programme-level "investigations"

The clinical tests for an obstetric emergency are in section 4 (Steps 3 and 4). At the **programme** level, these are the "investigations" you use to judge a service:

Data source	What you are looking for	Why
Partograph / Labour Care Guide audit	Charts started, lines crossed and acted on	Prevents rupture and fistula
HMIS/DHIS2 data: births, complications, CS rate, case fatality	EmONC indicators	Monitor met need and quality
EmONC needs assessment (signal functions in last 3 months)	Whether facilities are truly functional	Plan where to invest
MPDSR reviews, verbal autopsies, near-miss audits	Causes and avoidable factors	Drive the response
Criterion-based audit (e.g. % eclampsia given $MgSO_4$; % women with AMTSL)	Quality of care against standards	Quality improvement
Stock and cold-chain records (oxytocin, $MgSO_4$, blood)	Stock-outs, broken cold chain	A signal function without supplies is not available

G. Complications of failed safe-motherhood systems

"Complications" here means the consequences of failure of safe-motherhood systems:

For the woman	For the baby and family	For the system
Death; **near miss** with long-term morbidity	**Stillbirth** (intrapartum stillbirths are largely preventable)	Loss of trust leading to home births (a vicious cycle of the first delay)
Obstetric fistula (VVF/RVF) after neglected obstructed labour	Neonatal death, asphyxia, cerebral palsy	Cost of catastrophic care
Hysterectomy, infertility	**Orphaned children**: babies whose mothers die have much higher mortality	Health-worker burnout, moral injury
Sheehan's syndrome, renal failure	Household poverty, school dropout of older girls	Litigation, poor morale
Post-traumatic stress, depression		

7. Guidelines, Policy and Law in Uganda

> **UGANDA CONTEXT**
> This topic follows the **Uganda Clinical Guidelines (UCG) 2023**, the **Essential Maternal and Newborn Clinical Care Guidelines for Uganda (MoH 2022)**, the **national MPDSR guidelines (2017)**, the **UN EmONC monitoring handbook (2009)** and the **WHO MCPC manual**.

- **Levels of care**: HC III = BEmONC (7 signal functions); HC IV, general hospitals and above = CEmONC (all 9). Emergencies go to the nearest facility able to perform the needed signal function (section 6B).
- **MPDSR**: maternal and perinatal deaths are **notifiable** (since 2008); notify within **24 hours**, review at the facility committee ideally within **7 days**, **no name, no blame**, and every review ends in a **response**. Uganda publishes a **National Annual Maternal and Perinatal Death Report**.
- **Free maternity care at public facilities**: user fees were abolished in Ugandan public facilities in 2001; informal payments and "buy your own gloves" demands are a recognised cause of the third delay Verify *year of user-fee abolition and current MoH policy wording*.
- **Respectful maternity care** is part of the national quality standards and of the **Patients' Charter** (rights to information, consent, privacy, dignity and non-discrimination) Verify *current edition of the Uganda Patients' Charter*. Detaining women for unpaid bills is a recognised form of abuse.
- **Abortion**: the law is restrictive (Constitution Article 22(2) and the Penal Code Act), which drives unsafe abortion and stigma; **post-abortion care is legal, a BEmONC signal function (removal of retained products), and must be given without delay or judgement** Verify *current MoH standards and guidelines on reducing morbidity and mortality from unsafe abortion*.
- **SDG 3.1**: global MMR below 70 per 100,000 live births by 2030, with no country above 140.

UGANDA CONTEXT

Where Ugandan practice and WHO/international guidance differ:

- **ANC contacts**: WHO 2016 recommends **at least 8 contacts**; Uganda has historically used 4 focused visits and is moving to the WHO model Verify *the number of ANC contacts in the current Uganda ANC policy*.
- **$MgSO_4$ regimen**: Uganda uses the **Pritchard (IM) regimen** (4 g IV + 10 g IM, then 5 g IM 4-hourly); WHO accepts either Pritchard or the **Zuspan (IV) regimen** (4 g IV, then 1 g/h infusion) where an infusion can be monitored.
- **Signal functions**: exams still use the classic **7 + 2** (UN 2009); the 2025 Delphi consensus proposed 25 signal functions, not yet adopted as national policy.
- **Labour monitoring**: the modified WHO partograph is the Ugandan standard; the WHO **Labour Care Guide** (2020) is being introduced Verify *national roll-out status of the Labour Care Guide*.

8. Key Points

High-yield points

- **Every pregnancy faces risk**: complications cannot be predicted, so every woman needs timely access to EmONC.
- **Maternal death**: pregnant or within 42 days, related to or aggravated by pregnancy, not accidental; late maternal death 42 days to 1 year.
- **BEmONC = 7 signal functions; CEmONC = 7 + CS + blood transfusion**; a function counts only if performed in the last 3 months.
- **Three delays**: deciding, reaching, receiving care; give one intervention for each.
- **Rapid initial assessment before any history**: shock, bleeding, fits, fever, severe pain, imminent birth.
- **Stabilise before referral**: $MgSO_4$ loading dose, PPH bundle, first antibiotic dose, first dexamethasone dose, escort, note with vitals, drugs and times, phone ahead.
- **First 24 hours postpartum** are the most dangerous: observe every 15 minutes for 2 hours and keep her at least 24 hours.
- **UN indicators**: at least 5 EmONC facilities (1 comprehensive) per 500,000; met need 100%; CS 5–15%; direct obstetric case fatality below 1%.

- **MPDSR**: notify in 24 hours, review in 7 days, no name no blame, always a response.
- **Uganda MMR**: 336 (2016) → 189 (2022); SDG target below 70 by 2030.

Numbers & doses

Item	Value
Maternal death	Pregnant or within **42 days** of the end of pregnancy; late maternal death **42 days to 1 year**
MMR	Deaths per **100,000 live births**; Uganda **336 (2016) → 189 (2022)**
SDG 3.1	Global MMR **<70** by 2030; no country **>140**
Signal functions	**7 BEmONC**; **9 CEmONC** (+ CS + blood transfusion); performed in the **last 3 months**
EmONC availability	**≥5 facilities (≥1 comprehensive) per 500,000**
Women with a major complication	About **15%** of expected births
Met need for EmONC	Target **100%**
CS rate (population)	**5–15%** of all births
Direct obstetric case fatality rate	**<1%**
MPDSR	Notify within **24 h**; review ideally within **7 days**; no name, no blame
Postnatal stay after vaginal birth	At least **24 h**
Shock index	**≥0.9 abnormal; ≥1.4 urgent**
Pre-referral $MgSO_4$	**4 g IV (20%) over 5–10 min + 10 g IM (5 g each buttock)**; next 5 g IM due 4 h later
Pre-referral PPH	**Oxytocin 10 IU**, **TXA 1 g IV** within 3 h, fluids, NASG; misoprostol **800 µg SL** if no oxytocin
Pre-referral sepsis	**Ampicillin 2 g IV + gentamicin 5 mg/kg IV + metronidazole 500 mg IV**
Pre-referral preterm	**Dexamethasone 6 mg IM** (first of 4 doses, 12-hourly)
Emergency CS	Decision-to-delivery **within 30–60 min**
ANC (WHO 2016)	At least **8 contacts**, first in the first trimester
Newborn	Ventilate within the **Golden Minute**

Exam traps

- **"The HC III has a vacuum extractor, so it offers assisted vaginal delivery."** Only if it was **performed in the last 3 months**.
- **"CEmONC = BEmONC + CS."** It is **+ CS and blood transfusion**.
- **"Refer her quickly without treatment so she gets there sooner."** Wrong: **stabilise first** ($MgSO_4$, PPH bundle, antibiotics), then refer with an escort.
- **Road-traffic death in pregnancy = maternal death.** No: accidental deaths are **pregnancy-related**, not maternal.
- **Malaria death 3 weeks postpartum = not maternal.** It is an **indirect maternal death**.
- **"MMR = deaths per 100,000 women."** That is the maternal mortality **rate**; the **ratio** uses live births.
- **"MPDSR identifies who was at fault."** It is **no-name, no-blame**, and is incomplete without a **response**.
- **Triage by order of arrival.** Triage by **danger signs**: shock, bleeding and fits go first.

Questions seniors ask

Viva questions

1. *Define maternal death.* Death while pregnant or within 42 days, related to or aggravated by pregnancy, not accidental.
2. *List the BEmONC signal functions.* Antibiotics, uterotonics, anticonvulsants (all parenteral); MRP; removal of retained products; assisted vaginal delivery; neonatal resuscitation. **CEmONC adds CS and blood transfusion.**
3. *What are the three delays? Give an intervention for each.* Deciding (education and birth preparedness), reaching (transport, waiting homes), receiving care (staffing, supplies, drills, MPDSR).
4. *What is the UN recommendation on EmONC availability?* ≥5 EmONC facilities, including ≥1 CEmONC, per 500,000 population.
5. *What is MPDSR and its principles?* Continuous identification, notification and review of deaths **with a response**; no name, no blame; confidentiality.
6. *What is SDG 3.1?* Global MMR <70 per 100,000 live births by 2030; no country >140.
7. *Why is MMR a better measure than maternal mortality rate for obstetric care?* MMR measures risk **per pregnancy/birth** (quality of obstetric care); the rate also reflects fertility levels.
8. *What is the acceptable direct obstetric case fatality rate in EmONC facilities?* **<1%**.

On the ward round

A woman is brought in fitting. What do you do in the first minute? Shout for help, protect the airway (left lateral), oxygen, and give the $MgSO_4$ loading dose; then check BP, glucose and the fetal heart.

Which signal functions define a HC III and a HC IV? HC III: the 7 BEmONC functions. HC IV: all 9 (adds caesarean section and blood transfusion).

What must go with a woman you refer? An IV line and the stabilising treatment, a health worker with an emergency kit, a referral note with vital signs, drugs, doses and times (and the partograph), a relative who can donate blood, and a phone call to the receiving facility.

Analyse this death using the three delays. Delay 1 (deciding), delay 2 (reaching), delay 3 (receiving adequate care), each with patient/family, community and health-system factors, then one action per delay.

What happens after a maternal death in your unit? Notify within 24 hours, collect the records, review at the facility MPDSR committee (ideally within 7 days), agree the cause and avoidable factors, make SMART recommendations, implement and follow up.

How do you know whether a district's EmONC is working? Look at met need (target 100%), the CS rate (5–15%), the direct obstetric case fatality rate (<1%) and whether signal functions were actually performed.

Why keep a woman for 24 hours after a normal birth? Most PPH deaths happen in the first 24 hours, especially the first 2; she needs tone checks and observations.

What is a maternal near miss and why review it? A woman who nearly died but survived (e.g. shock, ≥5 units of blood, hysterectomy for bleeding, eclampsia with coma); near misses are commoner than deaths and she can tell you what went wrong.

Mnemonics

- **BEmONC signal functions, "AAA + MRP + RR + AVD + NR"**: **A**ntibiotics, **A**nticonvulsants, **A**nti-haemorrhagic uterotonics (all parenteral); **M**anual **R**emoval of **P**lacenta; **R**emoval of **R**etained products; **A**ssisted **V**aginal **D**elivery; **N**eonatal **R**esuscitation. **CEmONC = + "Cut and Carry blood"** (CS and transfusion).
- **Three delays: "Decide, Drive, Deliver"**: decide to seek care, drive (reach) the facility, deliver (receive) adequate care.
- **5 per 500,000, at least 1 comprehensive** ("five-for-half-a-million").
- **42 days** (maternal death) vs **1 year** (late maternal death).
- **MPDSR: "Notify in 1 day, review in 1 week, respond always."**

- **SDG 3.1: "70 by 2030."**

OSCE tips

- **Referral station**: verbalise **stabilisation before transfer**, write the **referral note with vital signs, drugs and times**, and **call the receiving facility**.
- **Birth-preparedness counselling station**: cover **place of birth, skilled attendant, transport, money, blood donor, companion, danger signs, and items (mama kit)**.
- **Triage station**: do the rapid initial assessment aloud and allocate staff by danger signs, not by order of arrival (worked example 1, section 4).
- **Maternal death analysis**: structure it by the three delays, with patient/family, community and health-system factors in each, then one SMART action per delay (worked example 2, section 4).

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SELF-TEST

Flashcards

26 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Define maternal death.	Death of a woman **while pregnant or within 42 days** of termination of pregnancy, irrespective of duration or site, from any cause **related to or aggravated by** the pregnancy or its management, but **not accidental or incidental** causes.
2	Define late maternal death.	Death from direct or indirect obstetric causes **more than 42 days but less than 1 year** after the end of pregnancy.
3	Maternal mortality ratio: definition and Uganda's latest figure.	Maternal deaths per **100,000 live births**. Uganda: **336 (UDHS 2016) → 189 (UDHS 2022)**.
4	What is SDG target 3.1?	Reduce the global MMR to **below 70 per 100,000 live births by 2030**, with **no country above 140**.
5	List the 7 BEmONC signal functions.	1. Parenteral **antibiotics** 2. Parenteral **uterotonics** 3. Parenteral **anticonvulsants** ($MgSO_4$) 4. **Manual removal of placenta** 5. **Removal of retained products** (MVA) 6. **Assisted vaginal delivery** (vacuum) 7. **Neonatal resuscitation** with bag and mask
6	Which two signal functions make a facility CEmONC?	**Surgery (caesarean section)** and **blood transfusion**, in addition to all 7 basic functions.
7	When is a signal function counted as 'available'?	When it has actually been **performed at least once in the previous 3 months**, not merely because the equipment exists.
8	UN recommended availability of EmONC facilities.	At least **5 EmONC facilities, including at least 1 comprehensive, per 500,000 population**.
9	The three delays model (authors, year, delays).	**Thaddeus & Maine, 1994** ('Too far to walk'): delay in **deciding to seek care**, in **reaching** a facility, and in **receiving adequate care** at the facility.
10	Target direct obstetric case fatality rate in EmONC facilities.	**Less than 1%** of women with major direct obstetric complications treated at the facility.
11	Acceptable population CS rate range (UN EmONC indicator).	**5–15%** of all births. Below 5% suggests unmet need; well above 15% suggests over-use.
12	What does MPDSR stand for, and what are its key steps?	**Maternal and Perinatal Death Surveillance and Response**: identify and **notify (within 24 h)** → **review** (ideally within 7 days) → analyse → **respond** (action plan) → monitor. Principle: **no name, no blame**.
13	Pillars of Safe Motherhood (Nairobi 1987).	**Family planning, antenatal care, clean/safe delivery, essential obstetric care**, on a foundation of **primary health care and equity for women**. Later additions: postnatal and post-abortion care.
14	Leading causes of maternal death in Uganda.	**Haemorrhage (PPH)**, **hypertensive disorders**, **sepsis**, **abortion complications**, obstructed labour/ruptured uterus, and **indirect** causes (malaria, HIV/TB, anaemia, heart disease).
15	Direct vs indirect obstetric death: one example of each.	**Direct**: from obstetric complications or their management (e.g. PPH, eclampsia, ruptured uterus). **Indirect**: pre-existing or new disease aggravated by pregnancy (e.g. **malaria, HIV**, cardiac disease).

#	Question	Answer
16	Levels of the Ugandan health system and their EmONC role.	VHT → HC II (no deliveries) → **HC III (BEmONC)** → **HC IV (CEmONC: theatre + blood)** → general hospital → RRH → national referral (e.g. Kawempe).
17	Pre-referral treatment for eclampsia at a HC III.	**$MgSO_4$ loading: 4 g IV (over 5–10 min) + 10 g IM** (5 g in each buttock), antihypertensive if BP ≥160/110, IV line, catheter, escort, and a referral note with doses and times.
18	What is a maternal near miss?	A woman who **nearly died but survived** a complication during pregnancy, childbirth or within 42 days of termination, identified by WHO organ-dysfunction criteria (e.g. shock, massive transfusion, hysterectomy, ICU).
19	Seven categories of disrespect and abuse in childbirth (Bowser & Hill).	Physical abuse, non-consented care, non-confidential care, non-dignified care, discrimination, abandonment/neglect, **detention in facilities**.
20	Components of birth preparedness and complication readiness.	Planned **place of birth** with a skilled attendant, **transport**, **money**, a **blood donor**, a **companion**, knowing **danger signs**, the items needed (**mama kit**), and who will decide in an emergency.
21	Minimum WHO postnatal stay after an uncomplicated vaginal birth.	**At least 24 hours** in the facility, because most maternal deaths (especially from PPH) occur within the first 24 hours.
22	Rapid initial assessment (RIA): what do you look for at the door?	**Airway/breathing** difficulty; **shock** (cold clammy, pulse >110, SBP <90, confused; SI ≥0.9); **heavy bleeding**; **convulsing or unconscious**; **dangerous fever**; **severe abdominal pain**; **imminent delivery**. Any 'yes' = **emergency**, treat before taking a history.
23	Pre-referral stabilisation by emergency.	**Eclampsia**: $MgSO_4$ 4 g IV + 10 g IM, BP control. **PPH**: oxytocin, TXA 1 g, fluids, NASG, compression. **Sepsis**: first dose of ampicillin + gentamicin + metronidazole. **Preterm**: first dexamethasone 6 mg IM. **Obstructed labour**: fluids, catheter, antibiotics, **no oxytocin**. Always: IV line, escort, note, phone ahead.
24	Contents of a good referral note.	Identity; history; findings **with vital signs and times**; **drugs given with dose and time** (e.g. when the next $MgSO_4$ dose is due); reason for referral; the partograph; the referring worker's name and **phone number**.
25	How do you calculate the direct obstetric case fatality rate, and what is the target?	Deaths from direct obstetric causes in EmONC facilities ÷ women with major direct complications treated there. Example: 40 ÷ 2,700 = **1.5%**. Target **<1%**; above it points to poor quality of care (third delay).
26	Structure of a maternal death review presentation.	1. **Summary of events** (timeline). 2. **Cause of death**: underlying cause, contributing factors; direct or indirect. 3. **Three-delays analysis**. 4. **Avoidable factors** (patient/community, health worker, facility, system). 5. **SMART response** with a responsible person and deadline. No name, no blame.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A HC III in Kyenjojo reports that in the last 3 months it has given parenteral antibiotics, oxytocin and $MgSO_4$, performed MVA and manual removal of placenta, and resuscitated newborns with bag and mask. It has a vacuum extractor, but no vacuum delivery has been performed in 8 months. How should it be classified?

A Comprehensive EmONC facility
B Less than basic EmONC facility
C Basic EmONC facility
D Fully functional basic facility because the equipment is present
E Referral hospital

2. A 24-year-old died of PPH after a home delivery. Her husband was away and her mother-in-law refused to let her go to the HC IV until he returned, 5 hours after bleeding began. Which delay does this represent, and what is the MOST appropriate intervention to address it?

A Delay 2; build a new road
B Delay 3; train midwives in the E-MOTIVE bundle
C Delay 3; stock more blood at the HC IV
D Delay 2; provide an ambulance
E Delay 1; community education on danger signs and birth preparedness, including who will decide in an emergency

3. A primigravida at 37 weeks has an eclamptic fit at a HC III that has no theatre. The nearest HC IV with CEmONC is 2 hours away by ambulance. Her BP is 168/112. What is the MOST appropriate action BEFORE transfer?

A Give $MgSO_4$ 4 g IV plus 10 g IM loading dose, an antihypertensive (e.g. hydralazine 5 mg IV or nifedipine 10 mg orally), secure IV access and catheterise, then transfer with a health worker
B Transfer immediately without treatment to avoid delay
C Give diazepam 10 mg IV and transfer
D Wait at the HC III for delivery before referral
E Give $MgSO_4$ only if she fits again

4. A district of 1,000,000 people has 3 HC IVs performing CS and blood transfusion, and 4 HC IIIs performing all 7 basic signal functions. According to the UN recommendation, which statement is correct? Advanced

A The district has too few comprehensive facilities
B The district needs at least 20 EmONC facilities
C It needs ≥10 EmONC facilities including ≥2 comprehensive; it meets the comprehensive requirement but falls short of the total
D HC IIIs cannot be counted as EmONC facilities
E Only hospitals count as comprehensive facilities

5. A woman dies at a regional referral hospital 26 hours after arriving in labour. She waited 5 hours for a CS because there was no anaesthetist on duty, and then 3 hours for blood after the operation. Which delay primarily contributed and what is the BEST system response?

A Delay 1; health education in her village
B Delay 2; provide transport vouchers
C No delay; the death was unavoidable
D Delay 3; fix anaesthetist call rotas and blood availability, identified and actioned through an MPDSR review
E Delay 2; build a maternity waiting home

6. A maternal death occurs at your HC IV on a Saturday night. According to Uganda's MPDSR guidelines, which is the CORRECT first step?

A Notify the district and MoH within 24 hours
B Wait for the monthly MPDSR meeting before reporting
C Identify the midwife responsible and suspend her
D Report only if the family complains
E Notify only if the death was due to a direct cause

7. Which of the following is NOT one of the seven basic EmONC signal functions?

A Parenteral anticonvulsants
B Manual removal of the placenta
C Assisted vaginal delivery
D Blood transfusion
E Neonatal resuscitation with bag and mask

8. A woman at 32 weeks dies from a road traffic accident while travelling to the market. How should this death be classified?

A Direct maternal death
B Indirect maternal death
C Late maternal death
D Maternal near miss
E Pregnancy-related death but not a maternal death (accidental/incidental cause)

9. A woman who is 3 weeks postpartum dies from cerebral malaria at a general hospital. How is this death BEST classified? Advanced

A Direct maternal death
B Indirect maternal death
C Not a maternal death because malaria is an infection
D Late maternal death
E Coincidental death

10. A district health officer wants a single indicator to judge whether women with major obstetric complications are actually receiving EmONC. Which indicator is MOST appropriate? Advanced

A Number of ANC visits per woman
B Contraceptive prevalence rate
C Met need for EmONC (proportion of expected complications treated in EmONC facilities)
D Number of TBAs trained
E Number of hospital beds

11. A 17-year-old is shouted at and slapped by a midwife during labour for 'not pushing properly', and is detained after delivery because her family cannot pay for the mama kit. Which statement BEST explains why this matters for maternal mortality?

A It only matters legally, not clinically
B It is acceptable to ensure cooperation in the second stage
C It primarily affects the third delay only
D Detention is permitted until bills are paid
E Disrespect and abuse deter women from facility birth in future, worsening the first delay, and violate their rights

12. At a MPDSR review of a PPH death, the committee finds that the woman delivered at 22:00, was not observed until 06:00, and was found in shock with an empty oxytocin cupboard. Which response is MOST in keeping with MPDSR principles?

A Dismiss the midwife on duty
B Record the cause as 'unavoidable PPH' and close the file
C Develop an action plan: postpartum observation protocol (every 15 min for 2 h, then hourly), a night staffing review and an oxytocin stock-management system, each with a responsible person and deadline
D Keep the findings confidential and take no action
E Report the case to the media to create pressure

13. A HC IV reports a caesarean section rate of 2% of all births in its catchment population, and a high rate of ruptured uteri and obstetric fistula. What does this MOST likely indicate? Advanced

A Excellent intrapartum care, so CS is not needed
B Unmet need for caesarean section (poor access to CEmONC)
C Over-use of CS
D Accurate data with no problem
E High use of vacuum extraction replacing CS

14. A woman arrives at a busy HC IV maternity unit. Which finding on rapid initial assessment requires IMMEDIATE emergency management before any history is taken?

A Convulsions
B Swollen feet at 36 weeks with normal BP
C Mild backache in early labour
D Request for family planning counselling
E Routine ANC visit at 28 weeks

15. A district wants to reduce maternal deaths from obstructed labour and PPH among women living on islands in Lake Victoria, more than 3 hours from the nearest HC IV. Which intervention MOST directly addresses the main barrier? Advanced

A Train more TBAs to conduct deliveries on the islands
B Increase the number of ANC visits to 8
C Distribute contraceptive pills only
D Establish a maternity waiting home near the HC IV and a water-ambulance referral system
E Stop deliveries at HC IIIs

Answers & explanations

1. B · Less than basic EmONC facility

B is right: a signal function must have been **performed within the last 3 months**. Missing **assisted vaginal delivery** makes this a 'less than basic' EmONC facility. A is wrong: CEmONC needs all 7 plus CS and blood transfusion. C and D are wrong: all 7 basic functions are required, and equipment alone doesn't count. E is wrong: a facility's level is not defined by signal functions in that way.

2. E · Delay 1; community education on danger signs and birth preparedness, including who will decide in an emergency

E is right: the problem was **deciding to seek care** (delay 1). Family decision-making, gender norms and lack of recognition of danger signs are addressed by community education, male involvement and **birth preparedness plans** that name the decision-maker. A and D address delay 2 (reaching care). B and C address delay 3 (care at the facility). All are useful, but not for the delay described.

3. A · Give $MgSO_4$ 4 g IV plus 10 g IM loading dose, an antihypertensive (e.g. hydralazine 5 mg IV or nifedipine 10 mg orally), secure IV access and catheterise, then transfer with a health worker

A is right: **stabilise before transfer**. The $MgSO_4$ loading dose is a BEmONC signal function and prevents further fits during transport; BP control reduces stroke risk; escort by a health worker and a referral note with doses and times. B is wrong: an untreated eclamptic can fit and aspirate or bleed into the brain in the ambulance. C is wrong: **$MgSO_4$ is superior to diazepam** for eclampsia. D is wrong: she needs a facility able to do CS if needed. E is wrong: $MgSO_4$ is given after the first fit to prevent recurrence.

4. C · It needs ≥10 EmONC facilities including ≥2 comprehensive; it meets the comprehensive requirement but falls short of the total

C is right: the standard is **≥5 EmONC facilities (≥1 comprehensive) per 500,000**, so for 1 million: **≥10 facilities including ≥2 CEmONC**. The district has 7 (3 of them CEmONC), so it meets the comprehensive requirement but not the total. A is wrong: 3 ≥ 2. B is wrong: 20 is a miscalculation. D is wrong: HC IIIs performing all 7 functions count as BEmONC. E is wrong: any facility performing all 9 functions, including a HC IV, is comprehensive.

5. D · Delay 3; fix anaesthetist call rotas and blood availability, identified and actioned through an MPDSR review

D is right: she reached the hospital, but care was delayed there. That is **delay 3** (receiving adequate care). The MPDSR response should target staffing (anaesthesia on-call) and **blood supply**. A, B and E address delays 1 and 2, which weren't the problem. C is wrong: a death with a 5-hour decision-to-delivery interval and a 3-hour wait for blood is clearly **avoidable**.

6. A · Notify the district and MoH within 24 hours

A is right: maternal deaths are **notifiable within 24 hours**, followed by a facility review (ideally within 7 days). B is wrong: it delays notification. C is wrong: MPDSR is **no-name, no-blame**, and punitive responses discourage honest reporting. D and E are wrong: **all** maternal deaths (direct and indirect) are notifiable.

7. D · Blood transfusion

D is right: **blood transfusion** (with CS) is one of the two extra **comprehensive** signal functions. A, B, C and E are all basic signal functions, along with parenteral antibiotics, parenteral uterotonics and removal of retained products.

8. E · Pregnancy-related death but not a maternal death (accidental/incidental cause)

E is right: deaths from **accidental or incidental** causes are excluded from maternal deaths, but they count as **pregnancy-related deaths** (any cause while pregnant or within 42 days). A and B are wrong: those require a cause related to or aggravated by the pregnancy. C is wrong: late maternal death is 42 days to 1 year after the end of pregnancy. D is wrong: she did not survive.

9. B · Indirect maternal death

B is right: malaria is a disease **aggravated by the physiological effects of pregnancy** (reduced immunity in pregnancy and the puerperium). Death within 42 days makes it an **indirect maternal death**. A is wrong: direct deaths result from obstetric complications (e.g. PPH, sepsis of genital-tract origin). C and E are wrong: malaria is not incidental; pregnancy increases its severity. D is wrong: she died within 42 days.

10. C · Met need for EmONC (proportion of expected complications treated in EmONC facilities)

C is right: **met need for EmONC** compares women treated for major direct obstetric complications in EmONC facilities with the expected number (about 15% of births); the target is 100%. A, B, D and E are useful for other purposes but don't measure whether complications are being treated.

11. E · Disrespect and abuse deter women from facility birth in future, worsening the first delay, and violate their rights

E is right: **respectful maternity care** is a right and a quality standard. Women who experience abuse (physical abuse, non-dignified care, **detention**) often avoid facilities next time, feeding **delay 1** and home births. A and B are wrong: abuse is never acceptable and has clinical consequences. C is wrong: it affects all delays, most importantly the decision to seek care. D is wrong: detention for unpaid bills is a human-rights violation.

12. C · Develop an action plan: postpartum observation protocol (every 15 min for 2 h, then hourly), a night staffing review and an oxytocin stock-management system, each with a responsible person and deadline

C is right: MPDSR must end in a **response** with **SMART actions** that address the system failures (monitoring, staffing, supplies). A is wrong: blame is counter-productive and violates the no-blame principle. B is wrong: the death was avoidable. D is wrong: confidentiality applies to identities, but **action** is mandatory. E is wrong: it breaches confidentiality.

13. B · Unmet need for caesarean section (poor access to CEmONC)

B is right: a population CS rate **below 5%** indicates **unmet obstetric need**. Women with obstructed labour are not getting timely surgery, leading to ruptures and fistulae. A is wrong: the complications show the opposite. C is wrong: over-use would be well above 15%. D is wrong: the associated morbidity shows a real problem. E is wrong: vacuum cannot replace CS for obstruction, and rupture/fistula rates would not be high.

14. A · Convulsions

A is right: RIA danger signs (unconscious/**convulsing**, shock, heavy bleeding, severe breathing difficulty, high fever, severe abdominal pain, imminent delivery) require immediate care. Convulsions mean eclampsia until proven otherwise: protect the airway and give $MgSO_4$. B, C, D and E are non-urgent and can wait in the normal queue.

15. D · Establish a maternity waiting home near the HC IV and a water-ambulance referral system

D is right: the main barrier is **geographic (delay 2)**. **Maternity waiting homes** (women stay near a CEmONC facility from about 36–38 weeks) and a **functional transport/referral system** (water ambulance) reduce it. A is wrong: TBAs cannot perform signal functions and are not skilled birth attendants. B is wrong: ANC is important but doesn't solve access in labour emergencies. C is wrong: FP helps but is not sufficient alone. E is wrong: HC IIIs provide BEmONC, and closing them worsens access.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Analysing a maternal death using the three delays

You are asked to present at the Mengo MPDSR meeting. Mrs T, a 28-year-old para 4 from a village in Mpigi, delivered at home with a TBA at 23:00. She began bleeding heavily after the placenta. The family waited for her husband to return (02:00), then spent 1.5 hours finding a boda-boda and money. At the HC III (04:00), the only midwife gave oxytocin 10 IU IM and referred her without an IV line. She reached Mengo at 06:30 in shock and died at 07:15 while blood was being cross-matched.

1. What is the medical cause of death and how will you classify it?

> **Primary postpartum haemorrhage** (probably atonic; para 4, home birth without AMTSL) leading to **hypovolaemic shock**. A **direct maternal death**.

2. Identify the delays and contributing factors.

> **Delay 1 (deciding)**: home birth with a TBA (birth preparedness failure), waiting 3 hours for the husband (gender and decision-making norms), not recognising the danger sign. **Delay 2 (reaching)**: no transport at night, 1.5 h to find money and a boda-boda, then HC III to Mengo transfer. **Delay 3 (receiving care)**: at the HC III, **no IV fluids, no TXA, no bimanual compression, no NASG, no escort**, and no call ahead; at Mengo, delay in getting blood.

3. Which interventions could have prevented this death? Give at least one per delay.

> Delay 1: **birth preparedness** counselling at ANC (place of birth, decision-maker, money, blood donor), **community misoprostol** for women likely to deliver at home, VHT education on danger signs, male involvement. Delay 2: **community emergency transport** scheme or ambulance numbers, savings groups, maternity waiting home. Delay 3: HC III **E-MOTIVE training and drills** (massage, oxytocin, TXA, fluids, examination, escalation), pre-referral stabilisation (**two IV lines, NASG**, bimanual/aortic compression during transfer), **phoning ahead**, and **emergency O-negative blood** available on arrival at the hospital.

4. How would you structure the MPDSR response?

> SMART action plan, e.g.:
>
> 1. Train all HC III midwives in PPH drills and pre-referral stabilisation within 3 months (responsible: district MCH focal person).
> 2. Pre-position NASG and TXA at all HC IIIs (DHO/medical stores, 1 month).
> 3. Establish a referral hotline and call-ahead protocol with Mengo (hospital administrator).
> 4. Hospital: an emergency blood protocol (O-negative on the labour ward) (blood bank lead).
> 5. Community birth-preparedness campaign via VHTs (health educator). Follow up at the next meeting. Keep it **no name, no blame**.

Take-away: Nearly every maternal death has several delays. MPDSR turns the story into actions, and the review isn't finished until someone is responsible for a change.

Case 2: Eclampsia at a HC III: stabilise and refer

You are a medical student on community placement at a HC III in Kasese. A 17-year-old primigravida at about 36 weeks is carried in by relatives after a fit at home 1 hour ago. She is drowsy, BP 170/115, and has 3+ proteinuria. The HC III has $MgSO_4$, hydralazine, IV fluids and a phone, but no theatre. The HC IV is 1.5 hours away.

1. What will you do in the first 5 minutes?

> **Call for help**; **rapid initial assessment**; place her in the **left lateral position**, clear the airway (suction), oxygen if available; protect from injury; IV access; **$MgSO_4$ loading: 4 g IV (20%) over 5–10 min + 10 g IM (5 g 50% in each buttock with lidocaine)**; catheterise (urine output); check the fetal heart.

2. How will you control her BP, and what are your monitoring targets?

> **Hydralazine 5 mg IV slowly**, repeat every 20–30 min until BP <160/110 (target around 140–150/90–100), max 20 mg; or **nifedipine 10 mg orally** if she can swallow safely (repeat after 30 min). Monitor for $MgSO_4$ toxicity: **patellar reflexes present, RR ≥16/min, urine ≥30 mL/h**; have **calcium gluconate 1 g IV** ready.

3. What will you include in the referral?

> Call the HC IV to alert the team; a **referral note** with the history (time of fit), **vital signs with times**, **$MgSO_4$ and antihypertensive doses and times**, urine output, fetal heart, and the partograph if in labour. Transfer in an **ambulance**, **accompanied by a health worker** with $MgSO_4$ (next IM dose due 4 h after loading), calcium gluconate, a delivery kit and a bag and mask. Ask a relative who could donate blood to come.

4. At the HC IV she is stable but not in labour; cervix unfavourable. What is the definitive treatment?

> **Delivery** within about 12 hours of the fit, after stabilisation. **Induction** (misoprostol/oxytocin) if feasible with an unfavourable cervix and a quick expected course, or **CS** if induction isn't likely to succeed quickly, or for obstetric indications. Continue **$MgSO_4$ for 24 h** after delivery or the last fit; AMTSL (**oxytocin; avoid ergometrine**); close postpartum BP monitoring.

5. Which EmONC signal functions and which delays were involved?

> Signal functions: **parenteral anticonvulsant** (BEmONC at the HC III), **caesarean section** and possibly **blood transfusion** (CEmONC at the HC IV), and **neonatal resuscitation**. Delays: possible delay 1 (home fit, adolescent), delay 2 (1.5 h transfer), and delay 3 avoided by **pre-referral stabilisation**.

Take-away: A BEmONC facility saves lives by giving the signal function ($MgSO_4$) before referral, then transferring safely with a proper escort and referral note.

Case 3: Planning EmONC for a district

The District Health Officer of a rural district (population 500,000) asks your team for advice. Data: 2 HC IVs (one theatre functional, one closed for 6 months due to no anaesthetist), 12 HC IIIs (only 3 performed assisted vaginal delivery in the last 3 months), 20,000 expected births per year, 180 CSs last year, and 14 maternal deaths reported (9 from PPH).

1. How many functional EmONC facilities does the district have compared with the UN standard?

> UN standard for 500,000 people: **≥5 EmONC, ≥1 comprehensive**. Comprehensive: **1** (only one HC IV performs CS and, presumably, transfusion). Basic: only HC IIIs performing **all 7** functions count; at most **3** (those doing vacuum deliveries, if they do all others). So at most **4 of 5** needed. It meets the comprehensive minimum on paper, but with only one functioning theatre.

2. Calculate the CS rate and interpret it.

> 180 ÷ 20,000 = **0.9%**, far below **5%**, indicating a large **unmet need** for caesarean section. Women with obstructed labour, praevia and similar conditions are not reaching surgery (a delay 2 and 3 problem).

3. Estimate the met need for EmONC if 700 women were treated for major direct complications in EmONC facilities.

> Expected complications = 15% of 20,000 = **3,000**. Met need = 700 ÷ 3,000 = **about 23%**, far below the 100% target.

4. Recommend priority actions.

> 1. **Reopen the second HC IV theatre**: recruit or train anaesthetic officers, and ensure a blood supply (UBTS link, blood fridge).
> 2. Make **HC IIIs fully BEmONC**: vacuum delivery training and mentorship, supplies, drills.
> 3. **Referral system**: ambulance, call-ahead protocol, maternity waiting homes.
> 4. **PPH focus**: E-MOTIVE (calibrated drapes, TXA, oxytocin cold chain), 24-h postnatal stay.
> 5. **MPDSR** in all facilities, with district follow-up of actions.
> 6. Community: birth preparedness, FP, respectful care to increase facility use.

Take-away: EmONC indicators turn raw data into decisions: count what is actually performed, calculate the CS rate and met need, and fix the weakest link (often anaesthesia, blood and referral).

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.